Magnetism CP D1-D3

Thursday, November 30, 2023 9:22 PM

D1)[9] An overhead power line 8.5 meters above the ground carries 95A of current.

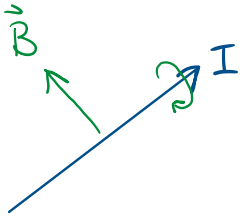
a) What is the magnitude of the magnetic field produced by the power line at ground level? How does it compare (percentage wise) to the Earth's magnetic field ($B_{Earth} \approx 50\mu T$)?
b) If the power line carries the current directly North-East (that is, 45° North of East), then which direction does the magnetic field vector at the ground point?

a) $B = \frac{\mu_0}{2\pi} \cdot \frac{I}{r} = 2\times10^{-7}\frac{N}{A^2} \cdot \frac{95\,A}{8.5\,m}$

$= 2.24\times10^{-6}\,T = \underline{2.24\,\mu T}$

$\frac{2.24\,\mu T}{50\,\mu T} \approx 4.5\%$ Earth mag field

b)

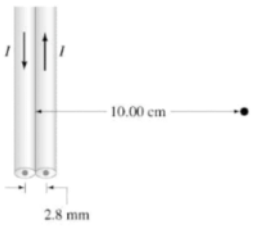


I points NE

B points NW

D2)[10] A long pair of wires conducts 25.0A of current to and from an instrument. The thin wires (dark grey) are separated by insulation (light grey) such that they are 2.8 mm apart. Consider a point 10 cm to the right of the midpoint between the two wires.

a) What is the total magnetic field at this point?
b) In which direction does the magnetic field point?



$\odot \vec{B}_L$
$\otimes \vec{B}_R$

a) From left wire:

$B = \frac{\mu_0}{} \quad \frac{25\,A}{} = 4.931\times10^{-5}\,T$

$$B_L = \frac{\mu_0}{2\pi} \cdot \frac{25\ A}{(0.1 + 0.0014)\ m} = 4.931 \times 10^{-5}\ T$$

(out of page)

From right wire:

$$B_R = \frac{\mu_0}{2\pi} \cdot \frac{25\ A}{(0.1 - 0.0014)\ m} = 5.071 \times 10^{-5}\ T$$

(into page)

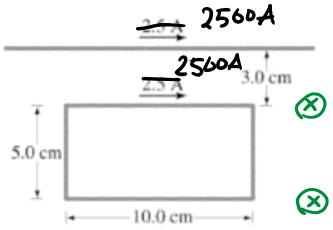
Total:

$$B_{tot} = (5.071 - 4.931) \times 10^{-5}\ T$$

$$= 1.4 \times 10^{-6}\ T = \underline{\underline{1.4\ \mu T}}$$

b) $\vec{B}_{tot}$ points "into page" since $B_R > B_L$

D3)[11] A long, straight power line hangs above the ground and carries 2.5 A of current to the right. A rectangular loop of wire is oriented vertically and parallel to the power line, as shown. There is a current of 2.5 A that flows clockwise in the loop.



a) What is the total magnetic force on the loop?

Hint: first find the magnetic field produced by the power line at the top and bottom of the loop, then find the force due to this field.

b) For what mass density of wire in grams per cm does the loop exactly "float" in the air?

a) Magnetic field at top of loop from power line

$$B_{top} = \frac{\mu_0}{2\pi} \cdot \frac{2500\,A}{0.03\,m} = 16.\bar{6}\,mT \quad (\text{into page})$$

So the force on the top section of wire is

$$F_{B,top} = l \cdot I \cdot B = 0.1\,m \cdot 2500\,A \cdot 16.\bar{6} \times 10^{-3}\,T$$

$$= 4.17\,N \quad (\text{direction "up"})$$

At bottom of loop,

$$B_{bottom} = \frac{\mu_0}{2\pi} \cdot \frac{2500\,A}{0.08\,m} = 6.25\,mT \quad (\text{into page})$$

$$F_{B,bottom} = l \cdot I \cdot B = 0.1 \cdot 2500 \cdot 6.25 \times 10^{-3}$$

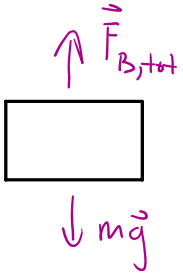
$$= 1.56\,N \quad (\text{direction "down"})$$

So total magnetic force is

$$F_{B,total} = F_{B,top} - F_{B,bottom} = (4.17 - 1.56)\,N$$

$$= \underline{\underline{2.61\,N}} \quad (\text{direction "up"})$$

b)



$$mg = 2.61\,N \rightarrow m = 0.266\,kg$$

$$= 266\,g$$

mass density (per unit length)

$$\frac{266\,g}{(5+10+5+10)\,cm} = \underline{\quad 8.87\,g/cm \quad}$$